

3D Localisation of a Wii Remote

Full Method and Derivations

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Problem Statement

Goal

Recover the 3-D position $\mathbf{P} = (p_x, p_y, p_z)^\top \in \mathbb{R}^3$ of a Wii remote in a fixed global frame $\mathcal{O}$, given:

- the orientation of the remote expressed as Euler angles (yaw ψ , pitch θ , roll ϕ);
- the pixel coordinates of two infrared beacons captured by the remote's on-board camera;
- the intrinsic parameters of that camera.

Known quantities

- Two beacons fixed in the global frame: $L_1 = (d, 0, 0)^\top$ and $L_2 = (-d, 0, 0)^\top$, separated by a known distance $2d$.
- Their projections on the camera sensor: $(u_1^{\text{raw}}, v_1^{\text{raw}})$ and $(u_2^{\text{raw}}, v_2^{\text{raw}})$.
- Camera intrinsics: image resolution (W, H) , horizontal field of view θ_w , vertical field of view θ_h .

Reference frames

Definition 1 (Global frame $\mathcal{O}$). *Fixed, world-attached frame with axes (X, Y, Z) . The beacons L_1, L_2 are given in this frame.*

Definition 2 (Remote frame $\mathcal{R}$). *Attached to the remote. Its origin coincides with the camera optical centre $\mathbf{P}$; its axes are the global axes rotated by the known Euler angles. All camera-model equations hold in this frame.*

Step 1 — Rotation Matrix from Euler Angles

Elementary rotation matrices

Three elementary rotations are defined about the Z , Y , and X axes respectively:

$$R_z(\psi) = \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \quad R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}. \quad (1)$$

ZYX composition (intrinsic yaw–pitch–roll)

Using the ZYX intrinsic convention (first yaw about Z , then pitch about the new Y , then roll about the new X), the global-to-remote rotation matrix is:

$$\mathbf{R} = R_z(\psi) R_y(\theta) R_x(\phi). \quad (2)$$

Explicit expanded form

Defining the shorthand $c_\alpha \equiv \cos \alpha$, $s_\alpha \equiv \sin \alpha$:

$$\mathbf{R} = \begin{pmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{pmatrix}. \quad (3)$$

Key property

Because $\mathbf{R}$ is orthogonal ($\mathbf{R}^\top \mathbf{R} = I_3$), its inverse equals its transpose:

$$\mathbf{R}^{-1} = \mathbf{R}^\top. \quad (4)$$

$\mathbf{R}$ transforms vectors from $\mathcal{O}$ to $\mathcal{R}$; $\mathbf{R}^\top$ transforms from $\mathcal{R}$ back to $\mathcal{O}$.

Step 2 — Camera Intrinsic Model

From FOV angles to pixel focal lengths

The pinhole camera geometry links the physical field of view to the focal length expressed in pixels. For the horizontal axis, half the image width subtends half the horizontal FOV:

$$\tan\left(\frac{\theta_w}{2}\right) = \frac{W/2}{f_x} \implies \boxed{f_x = \frac{W}{2 \tan(\theta_w/2)}}. \quad (5)$$

Likewise for the vertical axis:

$$\tan\left(\frac{\theta_h}{2}\right) = \frac{H/2}{f_y} \implies \boxed{f_y = \frac{H}{2 \tan(\theta_h/2)}}. \quad (6)$$

Remark 1. If $\theta_w/W = \theta_h/H$ the pixels are square and $f_x = f_y$. Otherwise the camera has a non-unit pixel aspect ratio.

Principal point

The optical axis intersects the sensor at its centre, called the *principal point*:

$$c_x = \frac{W}{2}, \quad c_y = \frac{H}{2}. \quad (7)$$

Intrinsic matrix

The four intrinsic parameters are assembled into the 3×3 upper triangular matrix:

$$K = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}. \quad (8)$$

Projection equation

A 3-D point $\mathbf{q} = (q_x, q_y, q_z)^\top$ in the remote frame $\mathcal{R}$ projects onto raw pixel coordinates via:

$$\begin{pmatrix} u^{\text{raw}} \\ v^{\text{raw}} \\ 1 \end{pmatrix} \sim K \begin{pmatrix} q_x/q_z \\ q_y/q_z \\ 1 \end{pmatrix}, \quad (9)$$

which gives explicitly:

$$u^{\text{raw}} = f_x \frac{q_x}{q_z} + c_x, \quad v^{\text{raw}} = f_y \frac{q_y}{q_z} + c_y. \quad (10)$$

Back-projection: pixel to ray direction

Inverting (10), a raw pixel $(u^{\text{raw}}, v^{\text{raw}})$ defines a *ray direction* in $\mathcal{R}$. Since K is upper triangular its inverse is:

$$K^{-1} = \begin{pmatrix} 1/f_x & 0 & -c_x/f_x \\ 0 & 1/f_y & -c_y/f_y \\ 0 & 0 & 1 \end{pmatrix}. \quad (11)$$

The normalised (homogeneous) direction vector for observation i is:

$$\mathbf{d}_i = K^{-1} \begin{pmatrix} u_i^{\text{raw}} \\ v_i^{\text{raw}} \\ 1 \end{pmatrix} = \begin{pmatrix} (u_i^{\text{raw}} - c_x)/f_x \\ (v_i^{\text{raw}} - c_y)/f_y \\ 1 \end{pmatrix}. \quad (12)$$

The beacon L_i lies somewhere along the ray $\{\lambda_i \mathbf{d}_i \mid \lambda_i > 0\}$ in frame $\mathcal{R}$.

Step 3 — World-Space Ray Directions

Expressing a world point in the remote frame

Let $\mathbf{P}$ be the (unknown) position of the remote in $\mathcal{O}$. The vector from the camera centre to beacon L_i , expressed in $\mathcal{O}$, is $(L_i - \mathbf{P})$. Rotating into $\mathcal{R}$:

$$\mathbf{q}_i = \mathbf{R}(L_i - \mathbf{P}). \quad (13)$$

Ray constraint

From the back-projection (12), $\mathbf{q}_i$ must be proportional to $\mathbf{d}_i$:

$$\mathbf{R}(L_i - \mathbf{P}) = \lambda_i \mathbf{d}_i, \quad \lambda_i > 0. \quad (14)$$

Rotating the ray back to the global frame

Multiplying both sides of (14) on the left by $\mathbf{R}^\top$:

$$L_i - \mathbf{P} = \lambda_i \mathbf{R}^\top \mathbf{d}_i. \quad (15)$$

Define the *world-space ray direction*:

$$\mathbf{r}_i = \mathbf{R}^\top \mathbf{d}_i \in \mathbb{R}^3. \quad (16)$$

$\mathbf{r}_i$ is fully known (depends only on the known rotation $\mathbf{R}$ and the observed pixel $\mathbf{d}_i$). The position satisfies:

$$\boxed{\mathbf{P} = L_i - \lambda_i \mathbf{r}_i, \quad i = 1, 2.} \quad (17)$$

Step 4 — Linear System for the Depths λ_1, λ_2

Equating the two position estimates

Equation (17) written for $i = 1$ and $i = 2$ must give the same $\mathbf{P}$:

$$L_1 - \lambda_1 \mathbf{r}_1 = L_2 - \lambda_2 \mathbf{r}_2. \quad (18)$$

Rearranging:

$$\lambda_1 \mathbf{r}_1 - \lambda_2 \mathbf{r}_2 = L_1 - L_2. \quad (19)$$

Right-hand side

With $L_1 = (d, 0, 0)^\top$ and $L_2 = (-d, 0, 0)^\top$:

$$L_1 - L_2 = \begin{pmatrix} 2d \\ 0 \\ 0 \end{pmatrix}. \quad (20)$$

Matrix form

Writing $\mathbf{r}_i = (r_x^i, r_y^i, r_z^i)^\top$, equation (19) becomes:

$$\underbrace{\begin{pmatrix} r_x^1 & -r_x^2 \\ r_y^1 & -r_y^2 \\ r_z^1 & -r_z^2 \end{pmatrix}}_{A \in \mathbb{R}^{3 \times 2}} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 2d \\ 0 \\ 0 \end{pmatrix}. \quad (21)$$

Least-squares solution

System (21) is *overdetermined* (3 equations, 2 unknowns). The least-squares solution minimises $\|A\boldsymbol{\lambda} - \mathbf{b}\|^2$:

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = (A^\top A)^{-1} A^\top \mathbf{b}, \quad \mathbf{b} = \begin{pmatrix} 2d \\ 0 \\ 0 \end{pmatrix}. \quad (22)$$

5.4.1 Explicit form of $A^\top A$

$$A^\top A = \begin{pmatrix} \|\mathbf{r}_1\|^2 & -\mathbf{r}_1 \cdot \mathbf{r}_2 \\ -\mathbf{r}_1 \cdot \mathbf{r}_2 & \|\mathbf{r}_2\|^2 \end{pmatrix}. \quad (23)$$

Its determinant is:

$$\det(A^\top A) = \|\mathbf{r}_1\|^2 \|\mathbf{r}_2\|^2 - (\mathbf{r}_1 \cdot \mathbf{r}_2)^2 = \|\mathbf{r}_1 \times \mathbf{r}_2\|^2. \quad (24)$$

This vanishes if and only if $\mathbf{r}_1 \parallel \mathbf{r}_2$, i.e. the two beacons project onto the same pixel — a physically degenerate configuration.

5.4.2 Explicit form of $A^\top \mathbf{b}$

$$A^\top \mathbf{b} = \begin{pmatrix} 2d r_x^1 \\ -2d r_x^2 \end{pmatrix}. \quad (25)$$

5.4.3 Closed-form solution

Let $\alpha = \|\mathbf{r}_1\|^2$, $\beta = \mathbf{r}_1 \cdot \mathbf{r}_2$, $\gamma = \|\mathbf{r}_2\|^2$, $\delta = \alpha\gamma - \beta^2$. Then:

$$\boxed{\lambda_1 = \frac{2d}{\delta}(\gamma r_x^1 + \beta r_x^2), \quad \lambda_2 = \frac{2d}{\delta}(\beta r_x^1 + \alpha r_x^2).} \quad (26)$$

Step 5 — Recovering the 3-D Position $\mathbf{P}$

Individual estimates

From (17):

$$\hat{\mathbf{P}}_1 = L_1 - \lambda_1 \mathbf{r}_1 = \begin{pmatrix} d \\ 0 \\ 0 \end{pmatrix} - \lambda_1 \mathbf{R}^\top \mathbf{d}_1, \quad (27)$$

$$\hat{\mathbf{P}}_2 = L_2 - \lambda_2 \mathbf{r}_2 = \begin{pmatrix} -d \\ 0 \\ 0 \end{pmatrix} - \lambda_2 \mathbf{R}^\top \mathbf{d}_2. \quad (28)$$

Final estimate

In the noiseless case $\hat{\mathbf{P}}_1 = \hat{\mathbf{P}}_2$. With noise, the average minimises the residual:

$$\boxed{\mathbf{P} = \frac{\hat{\mathbf{P}}_1 + \hat{\mathbf{P}}_2}{2}.} \quad (29)$$

Complete Algorithm

The full procedure is summarised below.

1. **Build the rotation matrix.** Compute $\mathbf{R} = R_z(\psi) R_y(\theta) R_x(\phi)$ using (3).

2. **Compute intrinsic parameters.**

$$f_x = \frac{W}{2 \tan(\theta_w/2)}, \quad f_y = \frac{H}{2 \tan(\theta_h/2)}, \quad c_x = \frac{W}{2}, \quad c_y = \frac{H}{2}.$$

3. **Back-project pixel observations.** For $i = 1, 2$, compute $\mathbf{d}_i$ via (12).

4. **Rotate rays to world frame.** $\mathbf{r}_i = \mathbf{R}^\top \mathbf{d}_i$ (equation (16)).

5. **Solve for depths.** Assemble A from (21) and solve (22) to obtain λ_1, λ_2 . The closed form is (26).

6. **Recover position.** $\mathbf{P} = \frac{1}{2}(L_1 - \lambda_1 \mathbf{r}_1 + L_2 - \lambda_2 \mathbf{r}_2)$.

Remarks and Edge Cases

Remark 2 (Degeneracy). *The system is degenerate when $\mathbf{r}_1 \parallel \mathbf{r}_2$ ($\delta = 0$). This happens only when the two beacons project onto identical pixels in the image, which is physically impossible if the bar is visible and the camera has finite resolution.*

Remark 3 (Sign of λ). *Both depths must be positive ($\lambda_i > 0$) for the beacons to be in front of the camera. A negative value indicates an inconsistent input (e.g. swapped beacon identities or an Euler-angle sign error).*

Remark 4 (Square pixels). *If $\theta_w/W = \theta_h/H$ then $f_x = f_y = f$ and the model reduces to the original single-parameter model with direction vectors $\mathbf{d}_i = ((u_i - c_x)/f, (v_i - c_y)/f, 1)^\top$.*

Remark 5 (Noise robustness). *The least-squares formulation in Step 4 automatically distributes pixel noise across all three coordinate equations. For higher accuracy under significant noise, one can iterate using the Gauss–Newton method on the full non-linear reprojection error.*